

Linux & System Security

week 1

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Users:

- step 1. Create User
2. Set Password
3. check
4. Switch User

```
[root@localhost ~]# useradd yuktaav
[root@localhost ~]# ls /home
yukta  yuktaav
[root@localhost ~]# passwd yuktaav
Changing password for user yuktaav.
New password:
BAD PASSWORD: The password is shorter than 8 characters
Retype new password:
passwd: all authentication tokens updated successfully.
[root@localhost ~]# groupadd skillset_go_edutech
[root@localhost ~]# tail /etc/group
cockpit-wsinstance:x:984:
gnome-initial-setup:x:983:
sshd:x:74:
chrony:x:982:
```

```
[yukta@localhost root]$
[yukta@localhost root]$ su yuktaav
Password:
[yuktaav@localhost root]$
```

Group :

5. Create Group

6. Add User to Group

7. Create File

8. Check permissions

9. now give permission to user

```
[root@localhost ~]# usermod -aG skillset_go_edutech yuktaav
[root@localhost ~]# groups yuktaav
yuktaav : yuktaav skillset_go_edutech
[root@localhost ~]# ls /home
yukta yuktaav
[root@localhost ~]#
[root@localhost ~]#
[root@localhost ~]#
```

```
[root@localhost ~]#
[root@localhost ~]# touch file.txt
[root@localhost ~]# ls -l file.txt
-rw-r--r--. 1 root root 0 Sep 25 21:56 file.txt
[root@localhost ~]# chmod 700 file.txt
[root@localhost ~]# ls -l file.txt
-rwx-----. 1 root root 0 Sep 25 21:56 file.txt
[root@localhost ~]# chmod x+ file.txt
chmod: invalid mode: 'x+'
Try 'chmod --help' for more information.
[root@localhost ~]# chmod +x file.txt
[root@localhost ~]# ls -l file.txt
-rwx--x--x. 1 root root 0 Sep 25 21:56 file.txt
[root@localhost ~]#
```

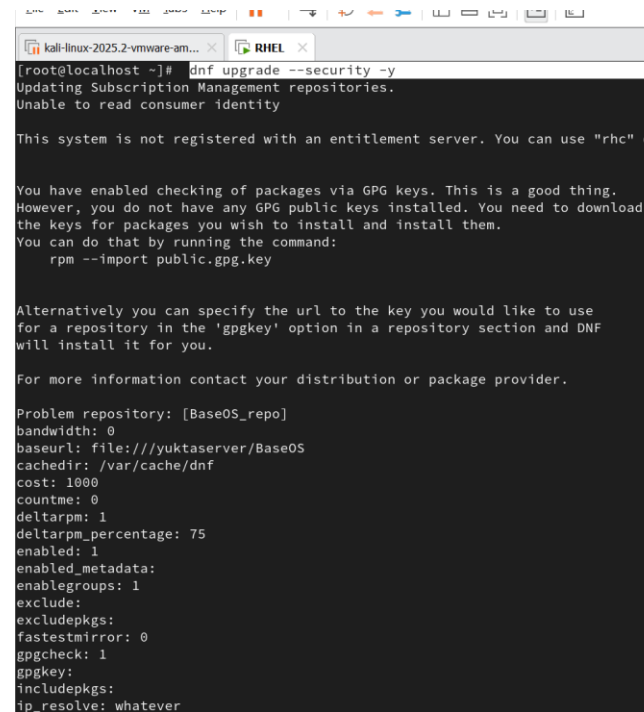
Change Ownership

```
Try 'chmod --help' for more information.  
[root@localhost ~]# chmod +x file.txt  
[root@localhost ~]# ls -l file.txt  
-rwx--x--x. 1 root root 0 Sep 25 21:56 file.txt  
[root@localhost ~]# chown yuktaav:skillset_go_edutech file.txt  
[root@localhost ~]# ls -l file.txt  
-rwx--x--x. 1 yuktaav skillset_go_edutech 0 Sep 25 21:56 file.txt  
[root@localhost ~]#  
[root@localhost ~]#  
[root@localhost ~]#  
[root@localhost ~]#
```

System Hardening and Security

Topic 1: System Updates and Cryptographic Policies

- To maintain system security and stability, first register the system with Red Hat Subscription Management
- Cmd - `dnf upgrade --security -y`



```
[root@localhost ~]# dnf upgrade --security -y
Updating Subscription Management repositories.
Unable to read consumer identity

This system is not registered with an entitlement server. You can use "rhc" or "subscription-manager" to register.

You have enabled checking of packages via GPG keys. This is a good thing.
However, you do not have any GPG public keys installed. You need to download
the keys for packages you wish to install and install them.
You can do that by running the command:
    rpm --import public.gpg.key

Alternatively you can specify the url to the key you would like to use
for a repository in the 'gpgkey' option in a repository section and DNF
will install it for you.

For more information contact your distribution or package provider.

Problem repository: [BaseOS_repo]
bandwidth: 0
baseurl: file:///yuktaserver/BaseOS
cachedir: /var/cache/dnf
cost: 1000
countme: 0
deltarpm: 1
deltarpm_percentage: 75
enabled: 1
enabled_metadata:
enablegroups: 1
exclude:
excludepkgs:
fastestmirror: 0
gpgcheck: 1
gpgkey:
includepkgs:
ip_resolve: whatever
```

To automate security updates, install and configure dnf-automatic

cmd - `dnf install dnf-automatic -y`

```
root@localhost ~]# dnf install dnf-automatic -y
Updating Subscription Management repositories.
Unable to read consumer identity

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For more information contact your distribution or package provider.

Problem repository: [BaseOS_repo]
bandwidth: 0
baseurl: file:///yuktaserver/BaseOS
cachedir: /var/cache/dnf
cost: 1000
countme: 0
```

```
mirrorlist:
module_hotfixes: 0
name: BaseOS repo
password:
priority: 99
protected_packages: dnf, dnf, redhat-release, setup, systemd, systemd-udev, grub2-tools-minimal, sudo, yum, grub2-efi-x64, dnf, redhat-release, setup, systemd, systemd-udev, grub2-tools-minimal, sudo, yum, grub2-efi-x64
proxy:
proxy_auth_method: any
proxy_password:
proxy_sslcert:
proxy_sslclientcert:
proxy_sslclientkey:
proxy_sslverify: 1
proxy_username:
repo_gpgcheck: 0
retries: 10
skip_if_unavailable: 0
sslcert:
sslclientcert:
sslclientkey:
sslverify: 1
sslverifystatus: 0
throttle: 0
timeout: 30
type:
user_agent: libdnf (Red Hat Enterprise Linux 9.5; generic; Linux.x86_64)
```

Cryptographic Policies

Policy	Level	Notes
DEFAULT	Balanced	TLS 1.2+, RSA-2048+, SHA-256+
FUTURE	High	TLS 1.3, RSA-3072+
FIPS	Highest	Suitable for regulated environments
LEGACY	Low	Supports older algorithms; not recommended

- Cryptographic policies is to control the strength and compatibility of security protocols and algorithms.
- These are 4 polices

Cryptographic Policies practical:

```
[root@localhost ~]# update-crypto-policies --show
DEFAULT
[root@localhost ~]# update-crypto-policies --set DEFAULT
Setting system policy to DEFAULT
Note: System-wide crypto policies are applied on application start-up.
It is recommended to restart the system for the change of policies
to fully take place.
[root@localhost ~]#
```

- To view the current cryptographic policy:
update-crypto-policies --show
- To set the system policy to DEFAULT:
update-crypto-policies --set DEFAULT

Firewall

- The firewalld service provides dynamic firewall management and allows administrators to define access control rules based on zones and conditions.
- to view the default zone:

```
[root@localhost ~]# firewall-cmd --get-default-zone  
public  
[root@localhost ~]#
```

To display the current configuration of the firewall:

```
[root@localhost ~]# firewall-cmd --list-all
public
  target: default
  icmp-block-inversion: no
  interfaces:
  sources:
  services: cockpit dhcpv6-client ssh
  ports:
  protocols:
  forward: yes
  masquerade: no
  forward-ports:
  source-ports:
  icmp-blocks:
  rich rules:
[root@localhost ~]#
```

SELinux

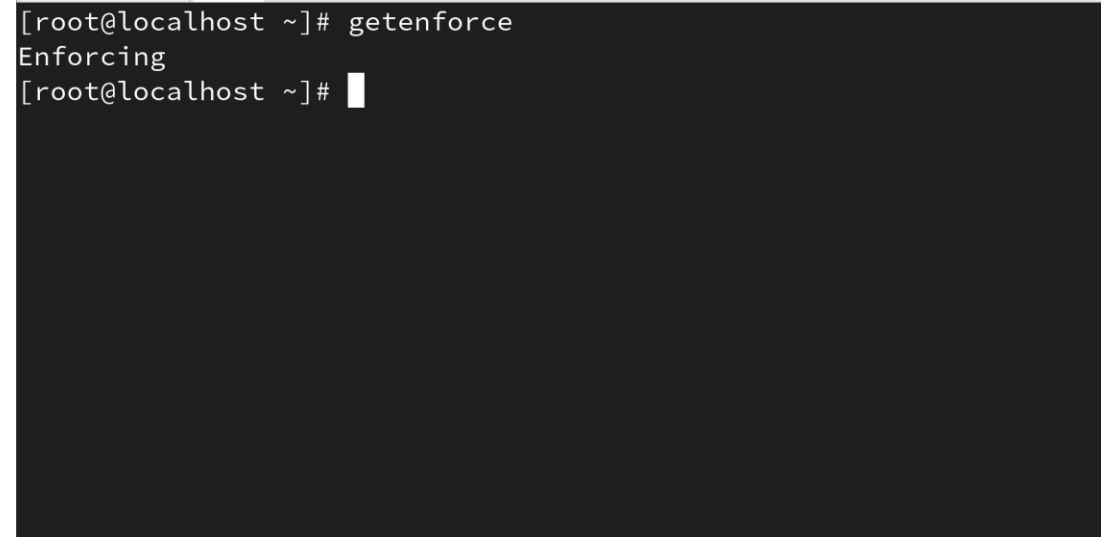
- **SELinux (Security-Enhanced Linux)** is a mandatory access control system that adds an additional layer of security beyond standard Linux permissions.
- There are **3 modes** :-
 1. ENFORCING :
 - Default in RHEL
 - Selinux enforcing the loaded polies
 2. PERMISSIVE:
 - Helpful for testing
 - Helpful in troubleshooting rule and application
 3. DISABLE:
 - In this mode selinux will be turned off

Selinux:

- To check the current SELinux mode:

- Cmd

getenforce

A terminal window with a dark background and light gray text. The prompt is [root@localhost ~]#. The command getenforce has been entered, and the output is Enforcing. The prompt is now [root@localhost ~]# with a white cursor bar.

```
[root@localhost ~]# getenforce
Enforcing
[root@localhost ~]#
```

To set SELinux to enforcing mode:
setenforce 0

```
[root@localhost ~]# getenforce
Enforcing
[root@localhost ~]# setenforce 0
[root@localhost ~]# getenforce
Permissive
[root@localhost ~]#
```

Key Principle

- firewalld controls which network packets are permitted to reach the system, while SELinux controls what processes are allowed to do after access is granted. Together, they provide a strong defense-in-depth security model.

- Resouces

Red Hat Enterprise Linux 9.0 RH134

Red Hat System Administration II

Edition 2